

Artificial Intelligence in the Knowledge Economy

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github.com/amchavezunyen/ai-05-ide

Why AI changes more than productivity

Before AI

Many workers handle routine cases. Scarce experts answer the difficult questions that workers cannot solve.

A hierarchy lets one expert's knowledge support many workers, but expert time remains scarce.

The AI shock

AI adds scalable knowledge. Its effect depends on how difficult a problem it can solve and whether it may produce on its own.

Those two features change who works with whom and how the team divides its output.

The paper's question. When AI enters the hierarchy, who gets a better expert, who captures the surplus, and does every unit of compute produce?

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Capability versus autonomy

Capability: z_{AI}

Which problems can the AI solve?

It determines whether AI is basic or advanced, whether firms use a nonautonomous AI, and whether autonomous AI creates winners at the bottom.

Autonomy: feasible roles

Can AI pursue production opportunities, or may it only advise humans?

It changes the AI's outside option, who captures team output, and whether available compute produces independently.

The trap. “Distribution depends on autonomy, not capability” collapses the model's two-dimensional taxonomy. Proposition 5 supplies a capability condition inside the autonomous regime.

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Model objects and the agent's problem

Primitives

$z \in [0, 1]$ human knowledge: fraction solved alone.

$x \sim U[0, 1]$ problem difficulty.

$h \in (0, 1)$ solver time consumed by one question.

$a \equiv z_{AI}$ AI capability.

μ available units of compute.

Equilibrium objects

$n(z)$ workers supervised by one solver.

$w(z), r$ human wage and compute rental price.

W, I, S human workers, independent producers, and solvers.

$m(z)$ solver knowledge matched to worker z .

W_a, S_a humans paired with AI on either layer.

Economic problem. Firms choose the organizational form, roles, and matches that maximize

$$\Pi = \mathbb{E}[\text{output}] - \text{human wages} - \text{compute cost},$$

subject to scarce solver time and market clearing. Humans choose the highest-paying occupation; compute owners rent to the highest bidder.

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One expert can support many workers

One worker, one opportunity

Knowledge $z \in [0, 1]$, difficulty $x \sim U[0, 1]$:

$$\Pr(x \leq z) = z.$$

A worker asks for help with probability $1 - z$. Each question consumes $h \in (0, 1)$ units of solver time.

Equilibrium sorting:

$$W \preceq I \preceq S, \quad m : W \rightarrow S \text{ strictly increasing.}$$

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One solver, several workers

$$h n(z)(1 - z) = 1 \implies \boxed{n(z) = \frac{1}{h(1 - z)}}.$$

A solver of knowledge $s \geq z$ raises success on every residual problem. More knowledgeable workers ask fewer questions, so they expand the solver's span of control.

A worker's wage reflects the match and the split

Human worker z , human solver s

$$\Pi_2^{nA}(s, z) = n(z)[s - w(z)] - w(s).$$

Zero profit gives

$$w(z) = s - \frac{w(s)}{n(z)}.$$

Worker pay equals match quality minus the solver's wage allocated across the team.

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Two sources of a wage change

$$\Delta w_L = \underbrace{(s_1 - s_0)}_{\text{match effect}} + \underbrace{\frac{q_0 - q_1}{n_L}}_{\text{share effect}}.$$

AI can worsen the worker's match while lowering the solver's claim on team output. The net sign need not be positive.

Autonomy gives AI a productive outside option

An AI agent has capability $a \equiv z_{AI}$ and uses one unit of compute. Abundant compute implies that some AI produces independently:

$$\Pi_1^{AI} = a - r^A = 0 \implies \boxed{r^A = a}.$$

AI as solver

$$n(z)[a - w^A(z)] - a = 0$$

$$\boxed{w^A(z) = a[1 - h(1 - z)]}.$$

AI as worker

$$n(a)[s - a] - w^A(s) = 0$$

$$\boxed{w^A(s) = n(a)(s - a)}.$$

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Capability decides where AI enters the hierarchy

Basic AI: $a \in \text{int } W$

$$W^* \subset W, \quad S^* \supset S.$$

AI is a relatively cheap routine worker. Demand for human solvers rises. Humans who remain workers receive worse solver matches.

Advanced AI: $a \in \text{int } S$

$$W^* \supset W, \quad S^* \subset S.$$

AI is a relatively cheap solver. Demand for human workers rises. The least knowledgeable workers can obtain a better match.

Propositions 3–4 connect occupations to wages. They determine the match effect that enters Proposition 5.

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Proposition 5: capability decides whether the bottom gains

Define winners below and above the AI's knowledge:

$$B = \{z \in [0, a] : w^A(z) > w(z)\}, \quad T = \{z \in [a, 1] : w^A(z) > w(z)\}.$$

Bottom

$$B \neq \emptyset \iff a > \bar{z}_{AI}, \quad \bar{z}_{AI} \in \text{int } W.$$

Basic AI can generate a negative match effect. The bottom gains only when capability is high enough for the share effect to dominate.

Key condition. The existence of bottom winners is a statement about AI capability, even though Proposition 5 already holds autonomy fixed.

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Top

$$T \neq \emptyset \quad \forall a \in [0, 1].$$

The positive share effect dominates under the maintained $h < h_0$. The source notes that this can fail for $h \geq h_0$ or $a = 1$.

A simple threshold explains the two numerical cases

For a bottom worker $z = 0$ directly assisted by autonomous AI,

$$w^A(0) = a(1 - h).$$

With $h = 1/2$ and the paper's pre-AI $w(0) \approx 0.35704365$,

$$w^A(0) > w(0) \iff a > \frac{w(0)}{1 - h} \approx \boxed{0.71408730}.$$

Capability a	Autonomous bottom wage $a/2$	Verdict
0.425	0.2125	bottom loses
0.850	0.4250	bottom gains

Boundary of the calculation. This is a two-type direct-assistance analogue. It illustrates the capability threshold but does not derive the continuum model's endogenous $\bar{z}_{AI}$.

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Proposition 6: restricting autonomy redistributes gains

Nonautonomous AI may advise humans but cannot pursue production opportunities. Abundant idle compute implies $r^N = 0$.

Activation condition

$a \leq w(0) \Rightarrow$ AI unused,

$a > w(0) \Rightarrow$ bottom types use AI.

A nonautonomous AI still needs enough capability to be worth using.

Comparisons

1. Autonomous output is strictly higher.
2. Nonautonomous AI creates some losers.
3. Near the bottom, nonautonomous wages are weakly highest.
4. Near the top, autonomous wages are weakly highest.

Strict wage statements use the proposition's activation and type conditions.

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Nonautonomy raises bottom pay but sacrifices output

Worker assisted by AI

$$n(z)[a - w^N(z)] - 0 = 0 \Rightarrow w^N(z) = a.$$

$$w^N(z) - w^A(z) = ah(1 - z) > 0.$$

Strict for $a > 0$, $h > 0$, $z < 1$, and economically active when $a > w(0)$.

Otherwise idle compute

Let Q_H be common human-related output and $k > 0$ remaining compute:

$$Q^N = Q_H, \quad Q^A = Q_H + ka.$$

$$Q^A - Q^N = ka > 0.$$

Nonautonomy removes the productive worker role.

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How to read the paper's results

First ask: how capable is AI?

If capability is too low, firms may not use a nonautonomous AI at all.

Within the autonomous regime, capability determines whether the least knowledgeable workers gain. The bottom needs AI to cross an endogenous threshold.

Then ask: what may AI do?

An autonomous AI can produce alone. That outside option raises the compute price and keeps spare compute productive.

A nonautonomous AI transfers more surplus to assisted workers, but leaves some compute idle and reduces total output.

Bottom line. A restriction on autonomy can help some low-knowledge workers while making the economy produce less. Capability and autonomy answer different parts of the distributional question.

INTERPRETATION. Economic reading of Propositions 5 and 6.

What I did analytically and computationally

Hand and algebra

- ▶ Derived $n(z) = 1/[h(1 - z)]$ from the solver's time constraint.
- ▶ Decomposed a worker wage change into match and share effects.
- ▶ Derived the discrete capability threshold.
- ▶ Derived the autonomy wage and output gaps.

Reproducible checks

- ▶ Recomputed $w(0) \approx 0.35704365$ from the paper's uniform example.
- ▶ Verified the 0.425 and 0.85 cases with a standard-library Python script.
- ▶ Formalized five discrete claims in Lean 4/Mathlib.
- ▶ Ran the paper-scoped AppliedModelingLib check before copying the Lean folder.

Claim boundary. None of these checks substitutes for the paper's continuum existence, matching, or neighborhood arguments.

Lean scope: a transparent discrete analogue

Lean verifies

- ▶ capability threshold equivalence;
- ▶ nonautonomous bottom-wage advantage;
- ▶ strict output gain from otherwise idle compute;
- ▶ match/share decomposition.
- ▶ positivity of span of control on the model domain.

Lean does not verify

- ▶ continuum equilibrium existence or uniqueness;
- ▶ occupational sets and matching maps;
- ▶ equality with the paper's endogenous threshold;
- ▶ the full top and bottom neighborhood claims.

Why this boundary matters. A short proof of an explicit algebraic analogue is stronger evidence than code that hides the equilibrium theorem in an assumption or a theorem named after the paper.

Required Lean slide: the capability threshold

1. Paper equation and discrete claim

$$w^A(z) = a[1 - h(1 - z)] \implies w^A(0) > w_0 \iff a > \frac{w_0}{1 - h}, \quad 0 < h < 1.$$

2. Lean statement

```
def bottomCapabilityThresholdSpec : Prop :=  
  ∀ a h w0 : ℝ,  
    0 < h -> h < 1 ->  
      autonomousAssistedWage a h 0 > w0 <->  
        a > capabilityThreshold w0 h
```

Most relevant proof step

```
have hden : 0 < 1 - h := sub_pos.mpr hh1  
unfold autonomousAssistedWage capabilityThreshold  
simp only [sub_zero, mul_one]  
constructor  
· exact fun hgain =>  
  (div_lt_iff0 hden).2 hgain  
· exact fun hcap =>  
  (div_lt_iff0 hden).1 hcap
```

3. What Lean represents

a : AI capability; h : helping cost; w_0 : pre-AI bottom wage. The strict domain $0 < h < 1$ makes $1 - h$ positive, so division preserves the inequality direction.

What Lean verifies

Exactly the equivalence in the displayed discrete model, including the nonzero positive denominator. It does not identify this threshold with the paper's continuum $\bar{z}_{AI}$.

Build/check: see

`lean/docs/CHECK_FAST_OUTPUT.txt`.

1. Hierarchy and capability threshold

- the primitives

worker knowledge : $(z < 1)$

solver knowledge : $(s \geq z)$

helping cost ($h \in (0,1)$)

- the span of control:

$$h n(z)(1-z) = 1 \rightarrow n(z) = \frac{1}{h(1-z)}$$

- All-human zero-profit condition

$$n(z)[s - w(z)] - w(s) = 0 \rightarrow w(z) = s - \frac{w(s)}{n(z)}$$

- subtract the pre- and post-AI worker wages and label the match

$$n(z)[s - w(z)] - w(s) = 0 \rightarrow w(z) = s - \frac{w(s)}{n(z)}$$

- abstract the pre- and post-AI worker wages and label the match and share effects

$$\Delta w_L = (s_1 - s_0) + \frac{q_0 - q_1}{n_L}$$

- set the bottom type to ($z=0$). with autonomous AI capability (a), use ($r^A = a$) and derive:

$$n(0)[a - w^A(0)] - a = 0 \rightarrow w^A(0) = a(1-h)$$

- derive and box the capability threshold:

$$w^A(0) > w_0 \iff a > \frac{w_0}{1-h}$$

- Insert ($h = 1/2$) and ($w_0 \approx 0.357$) check ($a = 0.425$) and ($a = 0.85$)

2. Autonomy comparison and verdict

- Nonautonomous AI cannot produce independently. With abundant idle compute write ($r^N = 0$)
- derive the assisted worker wage

$$n(z) [a - w^N(z)] - 0 = 0 \rightarrow w^N(z) = a$$

- compare with autonomous wage,

$$w^N(z) - w^A(z) = ah(1-z) > 0, \text{ with } \begin{matrix} (a > 0) \\ (h > 0) \end{matrix}$$

- let ($k > 0$) . derive:

$$\begin{matrix} (z < 1) \\ \text{and } (a > w(0)). \end{matrix}$$

$$Q^A - Q^N = ka > 0$$

- ∴ Capability determines whether the bottom gains within the autonomous regime. Autonomy changes the distribution of surplus and whether compute can produce independently. The 2 dimension cannot be collapsed into one.